

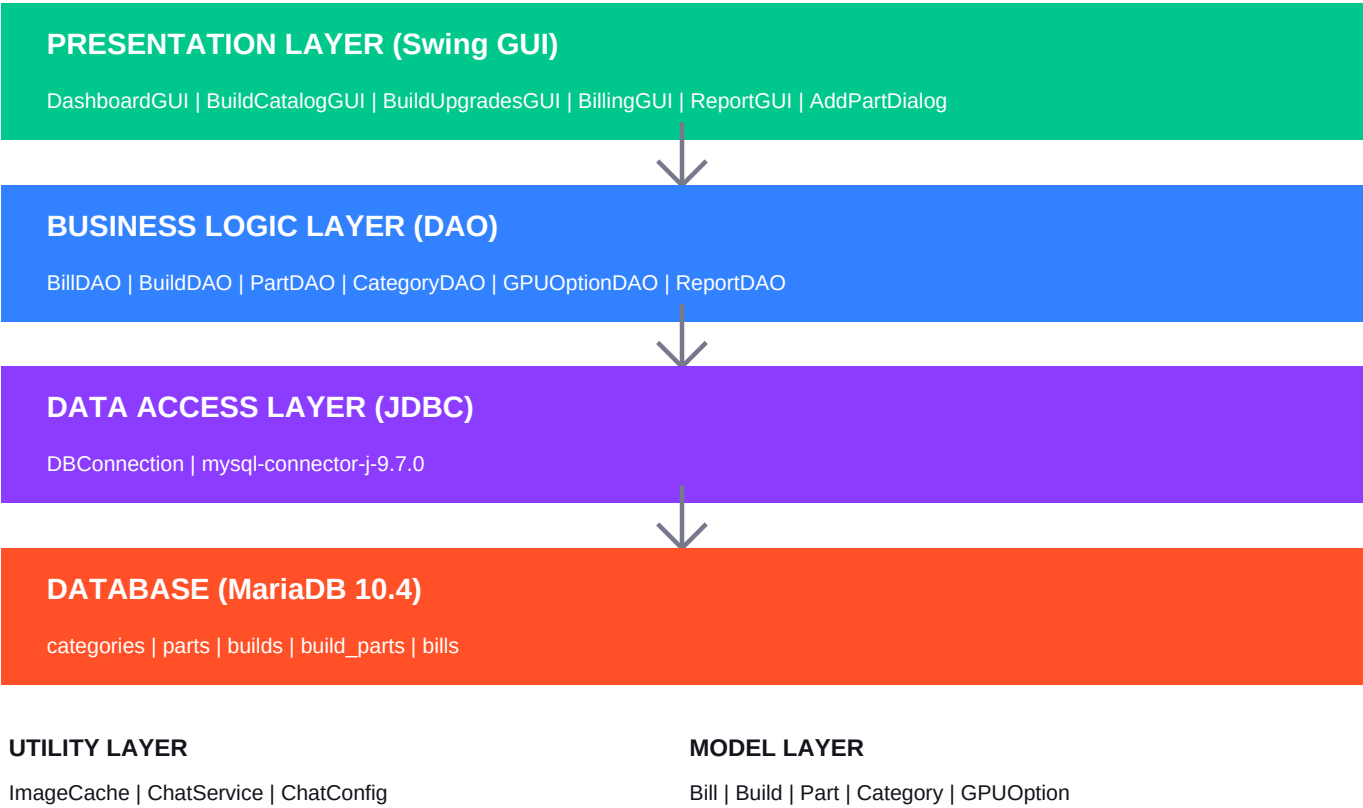
PC Build Store

System Architecture
and Design

Object Oriented Programming - Final Submission

2025-2026

1. Layered Architecture



Data Flow



2. Component Details

DashboardGUI

Main navigation hub. Displays hero banner, statistics cards (builds, parts, revenue, purchases), featured parts grid, and AI chat panel. Uses GridBagLayout for responsive design.

BuildCatalogGUI

6-slot PC builder with category tabs. Shows product grid with images, prices, and specs. Cart sidebar displays selected parts with running total.

BuildUpgradesGUI

Upgrade existing builds. Loads build from database, shows current parts in 6 slots, suggests compatible upgrades with compatibility checks.

BillingGUI

Purchase processing. Displays build summary, processes purchase (inserts into bills table), generates receipt with print and save options.

ReportGUI

Analytics dashboard. Revenue over time (line chart), parts distribution (pie chart), performance metrics (bar chart). All custom-painted.

ChatService

AI integration. SSE streaming via java.net.http.HttpClient. Parses both content and reasoning_content fields from KoboldCPP responses.

ImageCache

3-tier caching. LRU memory cache (120 entries), disk cache with timestamp validation, HTTP fetch with 8-thread parallel loading.

3. Design Patterns

Singleton

DBConnection provides single database connection point

DAO Pattern

Each entity has dedicated Data Access Object for CRUD

MVC

Separation of Model (POJOs), View (GUI), Controller (DAO)

Cache-Aside

ImageCache checks memory -> disk -> HTTP in order

Observer

Chat streaming callback processes chunks as they arrive